12.5 The Last Passenger's Seat

WORKSHEET

The question

STEP II 2021, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Pause and try Work out P_2 and P_3 by listing what can happen.

STAGE 1 The recurrence

Ask yourself Suppose T_1 sits in S_k, with $2 \leq k \leq n-1$. What happens to passengers $T_2, \ldots, T_{k-1}$? When T_k arrives, which seats are free, and how does the situation compare with a smaller train?

STAGE 2 The answer

Ask yourself $P_2 = P_3 = \frac{1}{2}$. Guess P_n and prove it by strong induction.

STAGE 3 The penultimate passenger

Ask yourself Try to find a slicker argument. At the end, which seat is left for T_n? Could it be S_2?

What's really going on

Symmetry arguments

The second proof is an example of a powerful idea in probability: if two outcomes are related by a symmetry of the random process, they have the same probability. The process here does not distinguish between S_1 and S_n until one of them is chosen, so they must be equally likely to be the survivor. Symmetry arguments often replace long calculations with a single observation. The next few sections are full of them: an extra coin toss in Section 12.6, a knockout draw in Section 12.7.

How many passengers are displaced?

The number of displaced passengers (those who do not get their own seat) is also nicely behaved. T_1 is displaced unless they choose S_1, and T_k (for $k \geq 2$) is displaced with probability $\frac{1}{n-k+2}$. Adding, the expected number of displaced passengers is

$$\left(1 - \frac{1}{n}\right) + \left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{2}\right) = H_n - \frac{1}{n},$$

roughly $\ln n$: even in a plane of 300 the chaos affects only about six passengers on average. The harmonic numbers, again (Section 9.11).

Try this next

1. **Two forgetful passengers.** If the first *two* passengers both choose at random, what is the probability that the last gets their own seat?
2. **Displaced passengers.** Verify the expected number of displaced passengers for $n = 3$ by listing the possibilities.
3. **An indirect route.** Show, using the recurrence, that $\sum_{r=2}^{n-1} P_r = \frac{n-2}{2}$ is consistent with the symmetry argument.

4 The last seat. Show that, conditional on the last passenger not getting their own seat, the last passenger gets S_1 with probability 1. Why?

Further reading

- **Peter Winkler, *Mathematical Puzzles: A Connoisseur's Collection*** (A K Peters). Includes the airline seat puzzle and many others solved by symmetry.
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover). A classic collection.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online).

The full guided solution is in Section 12.5:
https://beyondstep.keithmcnulty.org/ch12/airline-seat.html